END SEMESTER PRACTICAL EXAMINATION

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Part A: Data Acquisition and Understanding (10 Marks)

1. Domain and Business Problem

- **Domain:** Hospitality and Restaurant Analytics.
- **Business Problem:** The restaurant lacks visibility into its transactional data, making it difficult to optimize menu offerings and track revenue. The objective is to analyze historical transactions to identify top-performing product categories, track revenue trends over time, and evaluate payment method preferences to improve operational efficiency.

2. Dataset Source The dataset is sourced from Kaggle and is titled "Restaurant Sales". Specifically, it is the "Restaurant Sales Dataset with Dirt", which introduces realistic inconsistencies to simulate real-world data environments.

3. Key Tables and Fields The raw dataset is initially a flat file that is logically split into two main tables for preparation:

- **Order Table:** Contains transaction-level summary data. Key fields include Order ID, Order Total, Order Date, and Payment Method.

- **Customer Table:** Contains line-item details for each order. Key fields include Customer ID, Category, Item, Price, Quantity, and Order ID.

4. Suitability for Advanced Power BI Analysis

This dataset meets all the requirements for an advanced analytical solution:

- **Volume:** It contains 17,534 transactional rows, easily exceeding the 9,000-row minimum requirement.

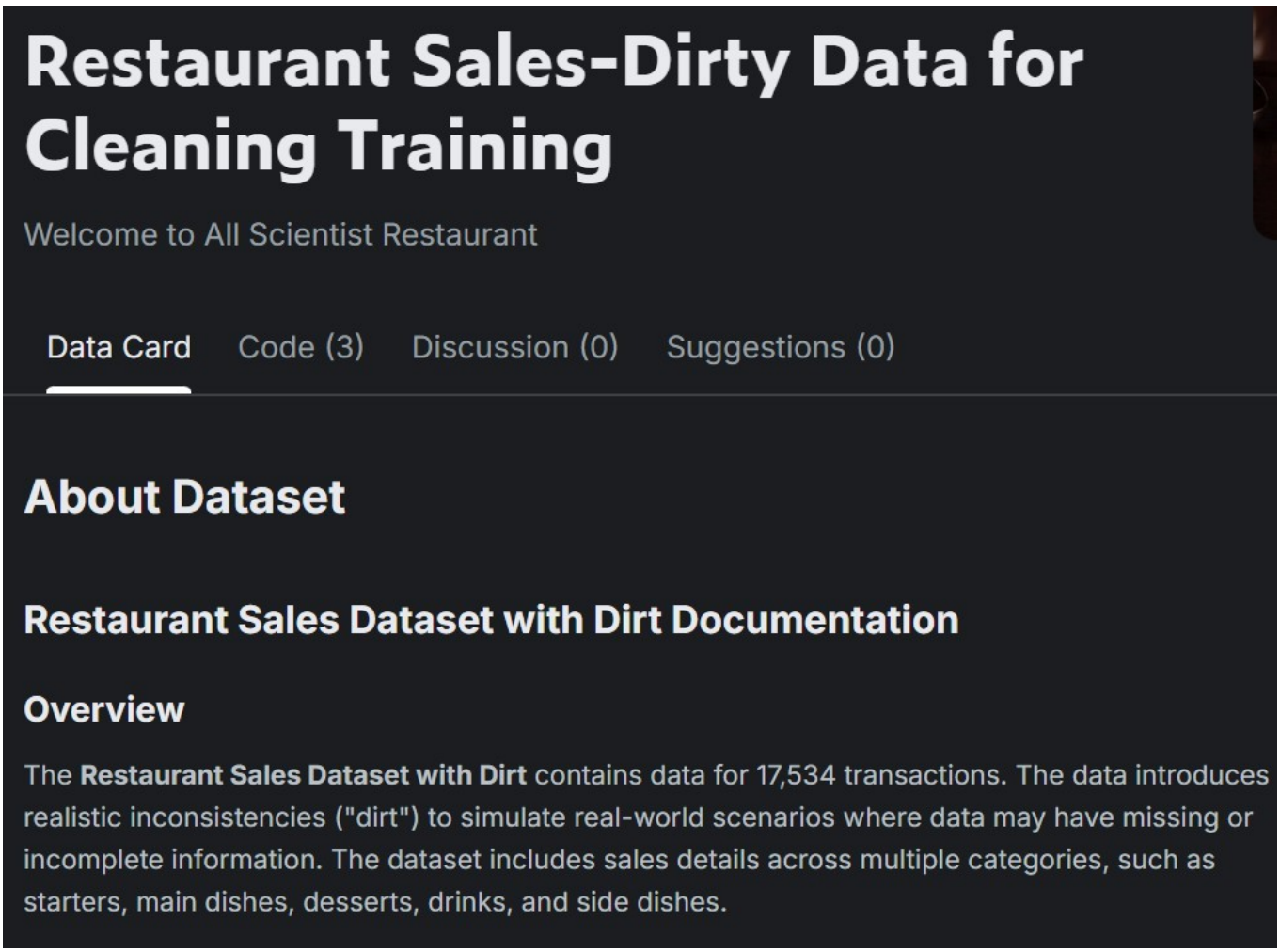
- **Complexity:** It includes a mix of categorical fields (Category, Payment Method) and numerical fields (Price, Quantity, Order Total) .

- **Time Intelligence:** It includes an 'Order Date' field, enabling calendar table creation and time-based DAX calculations .

- **Transformation Needs:** The "dirt" in the data (missing values, inconsistent text) requires substantial Power Query cleaning steps .

- **Relational Modeling:** The fields allow for normalization into a robust star schema consisting of a central Fact table and multiple dimension tables (Dim_Customer, Dim_Product, Dim_PaymentMethod, Dim_Date)

Details about the dataset can be found in the following image:



Restaurant Sales-Dirty Data for Cleaning Training

Welcome to All Scientist Restaurant

Data Card Code (3) Discussion (0) Suggestions (0)

About Dataset

Restaurant Sales Dataset with Dirt Documentation

Overview

The **Restaurant Sales Dataset with Dirt** contains data for 17,534 transactions. The data introduces realistic inconsistencies ("dirt") to simulate real-world scenarios where data may have missing or incomplete information. The dataset includes sales details across multiple categories, such as starters, main dishes, desserts, drinks, and side dishes.

Column Name	Description	Example Values
Order ID	A unique identifier for each order.	ORD_123456

Column Name	Description	Example Values
Customer ID	A unique identifier for each customer.	CUST_001
Category	The category of the purchased item.	Main Dishes, Drinks
Item	The name of the purchased item. May contain missing values due to data dirt.	Grilled Chicken, None
Price	The static price of the item. May contain missing values.	15.0,
None Quantity	The quantity of the purchased item. May contain missing values.	1, None
Order Total	The total price for the order (Price × Quantity). May contain missing values.	45.0, None
Order Date	The date when the order was placed. Always present.	2022-01-15
Payment Method	The payment method used for the transaction. May contain missing values.	Cash, None

A sample of the dataset is shown below:

Order ID A unique identifier for each order.	Customer ID A unique identifier for each customer.	Category The category of the purchased item.	Item The name of the purchased item.	Price The static price of the item.	Quantity The quantity of the purchased item.	Order Total The total price for the order (`Price * Quantity`)	Order Date The date when the order was placed.	Payment Method The payment method used for the transaction.
17534 unique values	CUST_066 1% CUST_100 1% Other (17127) 98%	Main Dishes 20% Starters 20% Other (10456) 60%	[null] 10% Pasta Alfredo 6% Other (14778) 84%	1 20	1 5	1 100	2022-01-01 2023-12-31	Credit Card 31% Cash 31% Other (6531) 37%
ORD_705844	CUST_092	Side Dishes	Side Salad	3.0	1.0	3.0	2023-12-21	Credit Card
ORD_338528	CUST_021	Side Dishes	Mashed Potatoes	4.0	3.0	12.0	2023-05-19	Digital Wallet
ORD_443849	CUST_029	Main Dishes	Grilled Chicken	15.0	4.0	60.0	2023-09-27	Credit Card
ORD_630508	CUST_075	Drinks			2.0	5.0	2022-08-09	Credit Card
ORD_648269	CUST_031	Main Dishes	Pasta Alfredo	12.0	4.0	48.0	2022-05-15	Cash
ORD_381680	CUST_031	Main Dishes	Salmon	18.0	5.0	90.0	2022-07-20	Digital Wallet
ORD_270994	CUST_071	Side Dishes	Garlic Bread	4.0	5.0	20.0	2022-08-19	Credit Card
ORD_146656	CUST_077	Main Dishes		15.0	3.0	45.0	2023-02-15	Cash
ORD_428611	CUST_083	Desserts		6.0	2.0	12.0	2023-12-16	Cash
ORD_743636	CUST_085	Main Dishes	Vegetarian Platter	14.0	5.0	70.0	2022-08-07	

To obtain multiple tables, I split the dataset into two tables: Order Table and Customer Table.

```
import pandas as pd import
numpy as np

# Load the dataset
data = pd.read_csv('restaurant_sales_data.csv')

# Splitting the data table1 =
data[
    ['Order ID', 'Order Total', 'Order Date', 'Payment Method']
]

table2 = data[
    ['Customer ID', 'Category', 'Item', 'Price', 'Quantity', 'Order ID']
```

```
# Save the tables to new CSV files
table1.to_csv('Order Table.csv', index=False)
table2.to_csv('Customer Table.csv', index=False)
```

The Order Table contains the following columns: 'Order ID', 'Order Total', 'Order Date', 'Payment Method'

Order ID	Order Total	Order Date	Payment Method
Missing: 0 (0%)	Missing: 430 (2%)	Missing: 0 (0%)	Missing: 1082 (6%)
Distinct: 17534 (100%)	Distinct: 41 (<1%)	Distinct: 730 (4%)	Distinct: 3 (<1%)
17534 Distinct values	Min 1.0 Max 100.0	730 Distinct values	Credit Card 31% Cash 31% Digital Wallet 31%

	Order ID	Order Total	Order Date	Payment Method
0	ORD_705844	3.0	2023-12-21	Credit Card
1	ORD_338528	12.0	2023-05-19	Digital Wallet
2	ORD_443849	60.0	2023-09-27	Credit Card
3	ORD_630508	5.0	2022-08-09	Credit Card
4	ORD_648269	48.0	2022-05-15	Cash
5	ORD_381680	90.0	2022-07-20	Digital Wallet
6	ORD_270994	20.0	2022-08-19	Credit Card
7	ORD_146656	45.0	2023-02-15	Cash
8	ORD_428611	12.0	2023-12-16	Cash
9	ORD_743636	70.0	2022-08-07	*Missing value*
10	ORD_117003	12.0	2023-12-09	Credit Card

The Customer Table contains the following columns: 'Customer ID', 'Category', 'Item', 'Price', 'Quantity', 'Order ID'

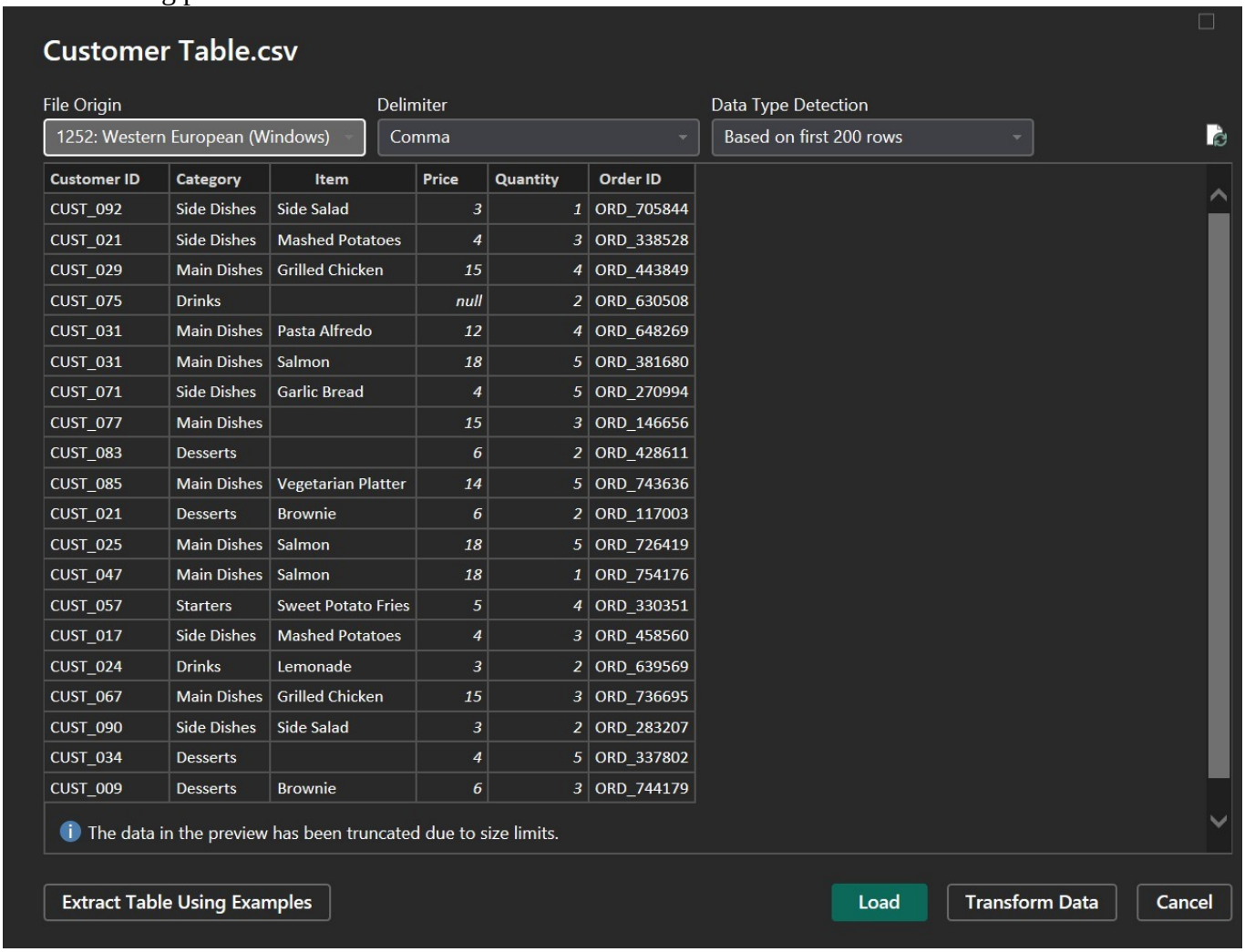
Customer ID	Category	Item	Price	Quantity	Order ID
Missing: 0 (0%)	Missing: 0 (0%)	Missing: 1758 (10%)	Missing: 876 (5%)	Missing: 430 (2%)	Missing: 0 (0
Distinct: 100 (<1%)	Distinct: 5 (<1%)	Distinct: 26 (<1%)	Distinct: 14 (<1%)	Distinct: 5 (<1%)	Distinct: 17534 (100
100 Distinct values	Main Dishes 20% Starters 20% Desserts 20% Other 40%	26 Distinct values	Min 1.0 Max 20.0	Min 1.0 Max 5.0	17534 Distinct values

	Customer ID	Category	Item	Price	Quantity	Order ID
0	CUST_092	Side Dishes	Side Salad	3.0	1.0	ORD_705844
1	CUST_021	Side Dishes	Mashed Potatoes	4.0	3.0	ORD_338528
2	CUST_029	Main Dishes	Grilled Chicken	15.0	4.0	ORD_443849
3	CUST_075	Drinks	*Missing value*	*Missing value*	2.0	ORD_630508
4	CUST_031	Main Dishes	Pasta Alfredo	12.0	4.0	ORD_648269
5	CUST_031	Main Dishes	Salmon	18.0	5.0	ORD_381680
6	CUST_071	Side Dishes	Garlic Bread	4.0	5.0	ORD_270994
7	CUST_077	Main Dishes	*Missing value*	15.0	3.0	ORD_146656
8	CUST_083	Desserts	*Missing value*	6.0	2.0	ORD_428611
9	CUST_085	Main Dishes	Vegetarian Platter	14.0	5.0	ORD_743636
10	CUST_021	Desserts	Brownie	6.0	2.0	ORD_117003
11	CUST_025	Main Dishes	Salmon	18.0	5.0	ORD_726419
12	CUST_047	Main Dishes	Salmon	18.0	1.0	ORD_754176

Part B: Data Cleaning and Transformation in Power Query (20 Marks)

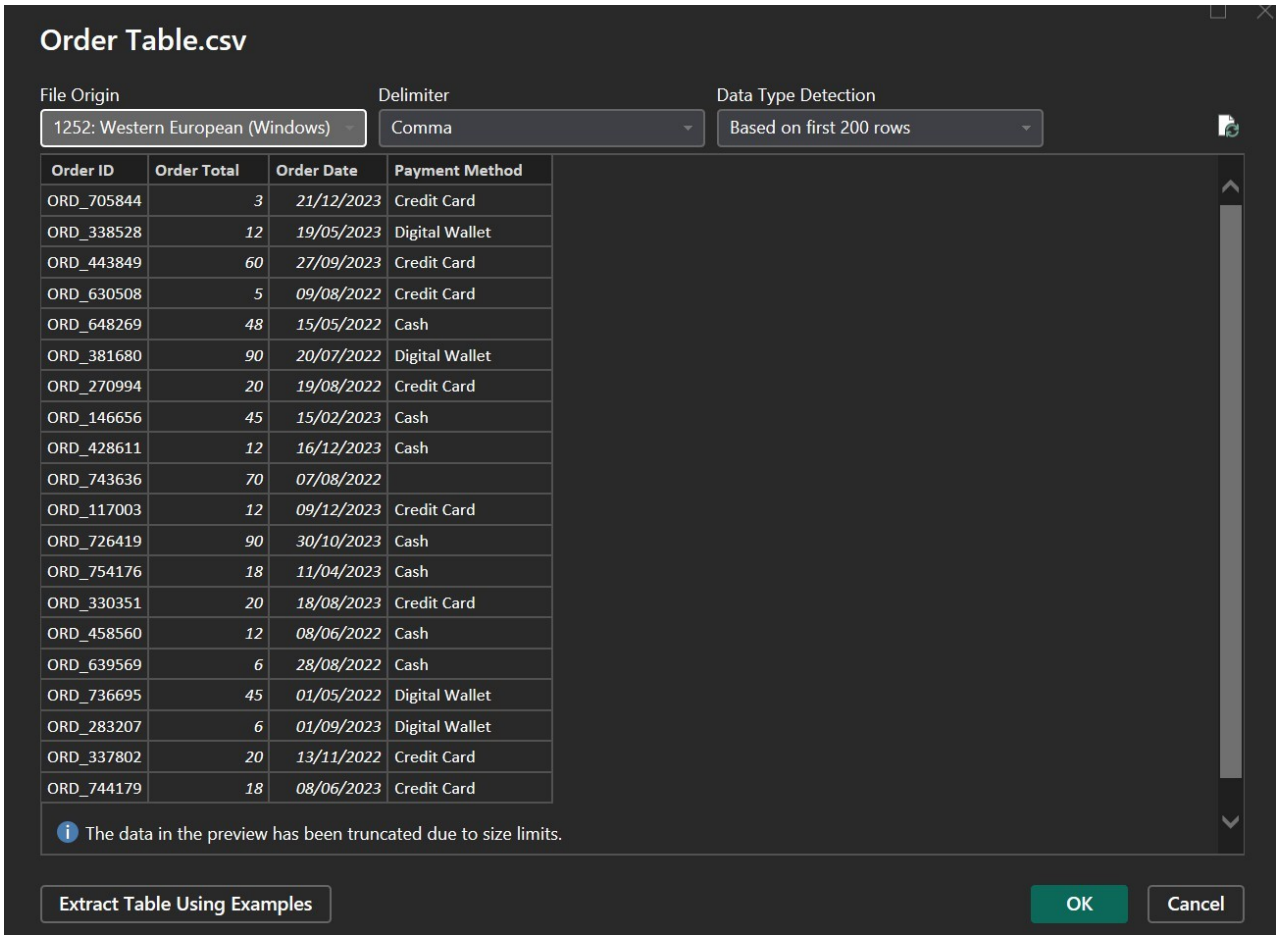
Loading the data in Power BI

- Loading process of the Customer Table



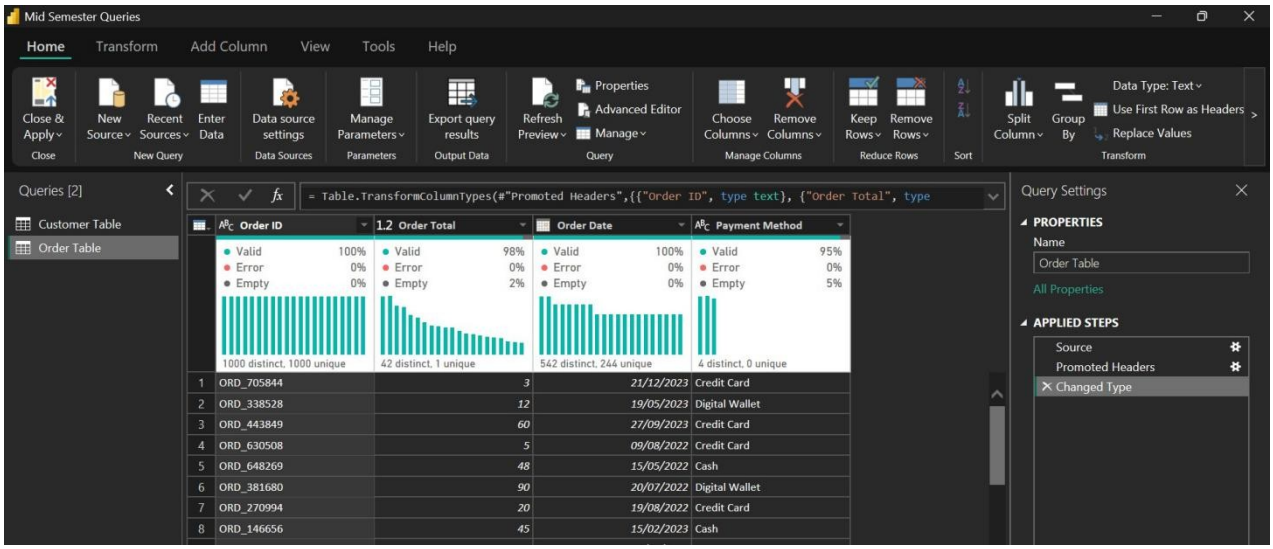
Customer ID	Category	Item	Price	Quantity	Order ID
CUST_092	Side Dishes	Side Salad	3	1	ORD_705844
CUST_021	Side Dishes	Mashed Potatoes	4	3	ORD_338528
CUST_029	Main Dishes	Grilled Chicken	15	4	ORD_443849
CUST_075	Drinks		null	2	ORD_630508
CUST_031	Main Dishes	Pasta Alfredo	12	4	ORD_648269
CUST_031	Main Dishes	Salmon	18	5	ORD_381680
CUST_071	Side Dishes	Garlic Bread	4	5	ORD_270994
CUST_077	Main Dishes		15	3	ORD_146656
CUST_083	Desserts		6	2	ORD_428611
CUST_085	Main Dishes	Vegetarian Platter	14	5	ORD_743636
CUST_021	Desserts	Brownie	6	2	ORD_117003
CUST_025	Main Dishes	Salmon	18	5	ORD_726419
CUST_047	Main Dishes	Salmon	18	1	ORD_754176
CUST_057	Starters	Sweet Potato Fries	5	4	ORD_330351
CUST_017	Side Dishes	Mashed Potatoes	4	3	ORD_458560
CUST_024	Drinks	Lemonade	3	2	ORD_639569
CUST_067	Main Dishes	Grilled Chicken	15	3	ORD_736695
CUST_090	Side Dishes	Side Salad	3	2	ORD_283207
CUST_034	Desserts		4	5	ORD_337802
CUST_009	Desserts	Brownie	6	3	ORD_744179

- Loading process of the Order Table



Order ID	Order Total	Order Date	Payment Method
ORD_705844	3	21/12/2023	Credit Card
ORD_338528	12	19/05/2023	Digital Wallet
ORD_443849	60	27/09/2023	Credit Card
ORD_630508	5	09/08/2022	Credit Card
ORD_648269	48	15/05/2022	Cash
ORD_381680	90	20/07/2022	Digital Wallet
ORD_270994	20	19/08/2022	Credit Card
ORD_146656	45	15/02/2023	Cash
ORD_428611	12	16/12/2023	Cash
ORD_743636	70	07/08/2022	
ORD_117003	12	09/12/2023	Credit Card
ORD_726419	90	30/10/2023	Cash
ORD_754176	18	11/04/2023	Cash
ORD_330351	20	18/08/2023	Credit Card
ORD_458560	12	08/06/2022	Cash
ORD_639569	6	28/08/2022	Cash
ORD_736695	45	01/05/2022	Digital Wallet
ORD_283207	6	01/09/2023	Digital Wallet
ORD_337802	20	13/11/2022	Credit Card
ORD_744179	18	08/06/2023	Credit Card

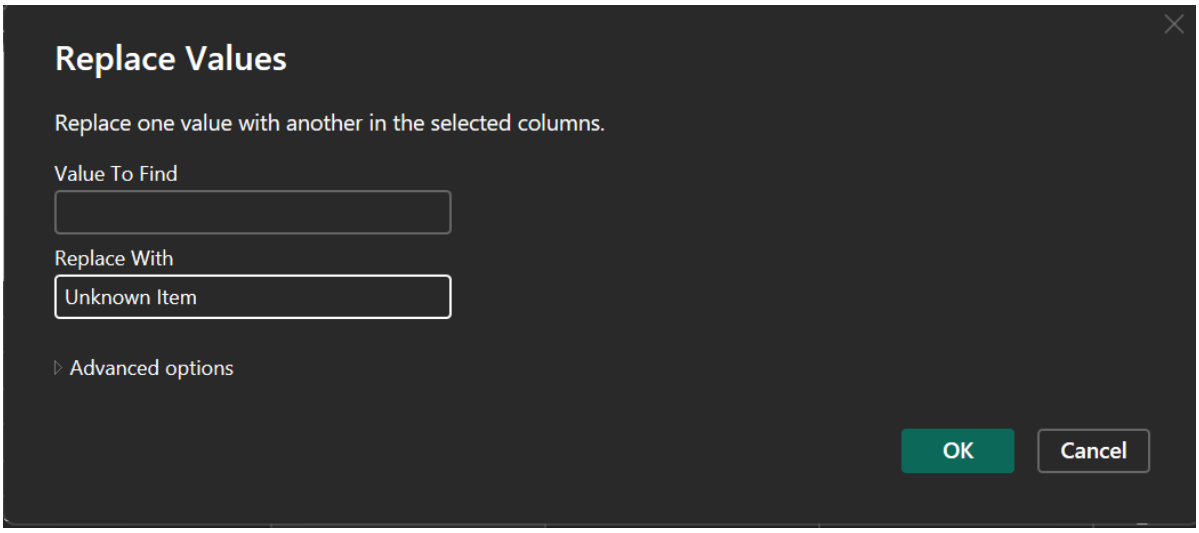
- Power Query Editor after loading the tables



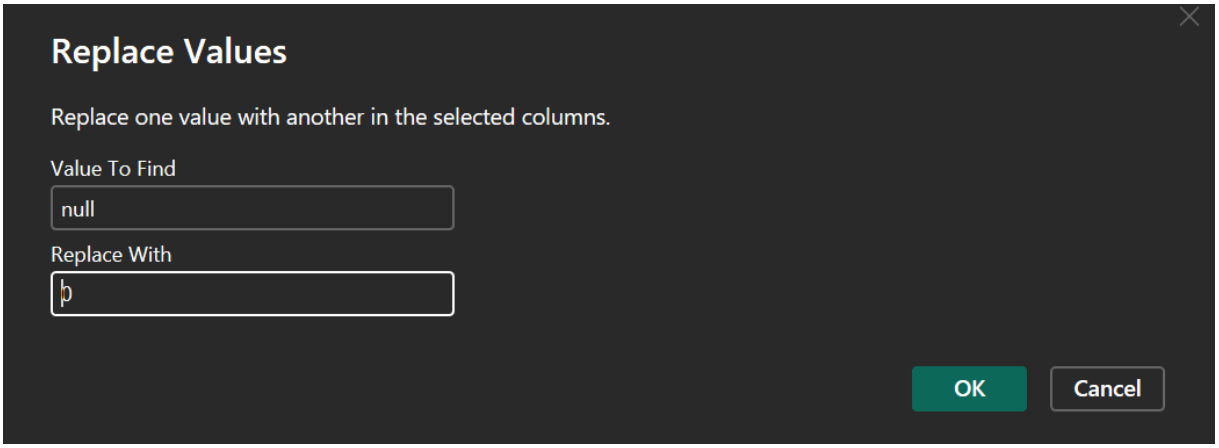
Data Cleaning and Transformation

- Removing duplicates in the Customer Table
- Handling missing or null values

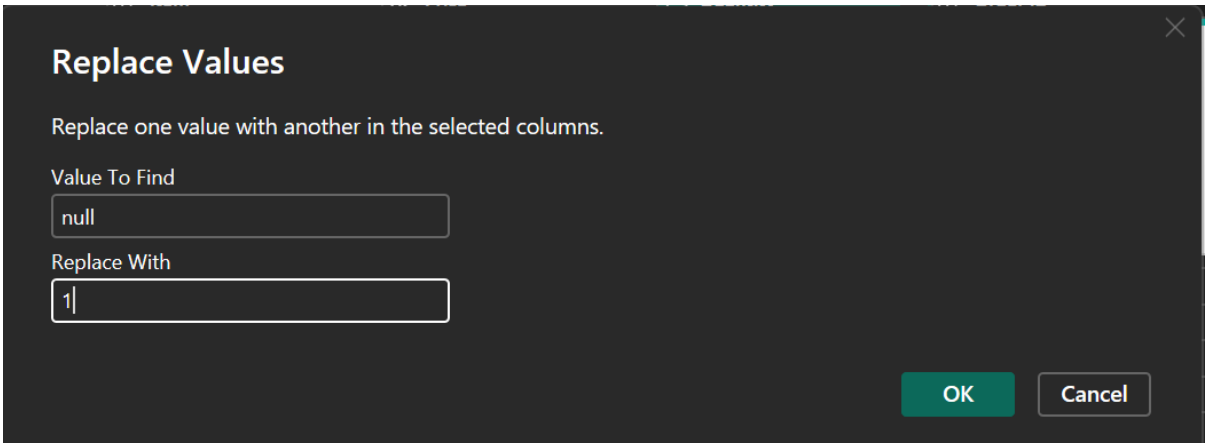
- Filled missing Items and Payment Method with "Unknown Item"



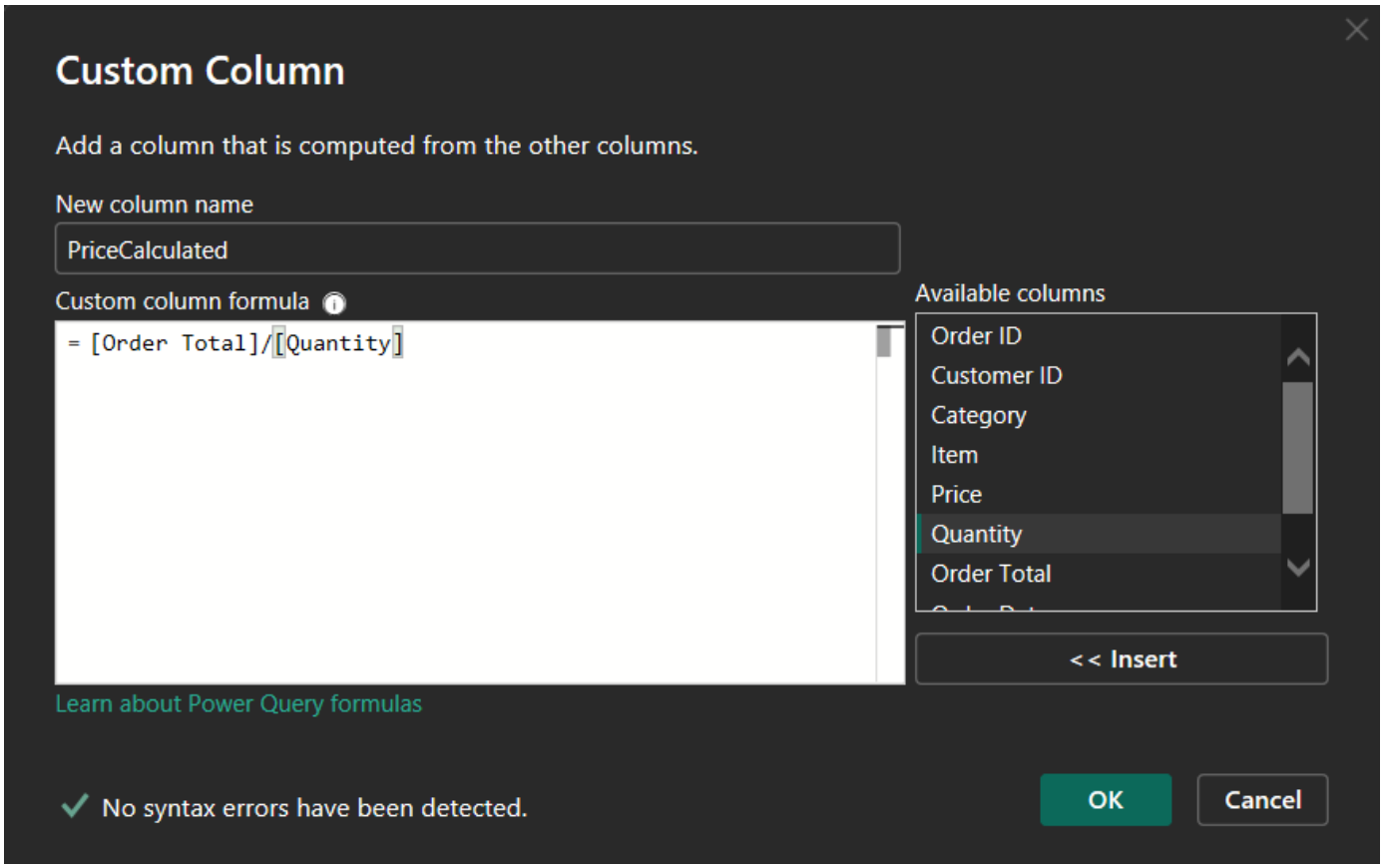
- Filling the missing Price:



- Filled missing Quantity with 1 because it is reasonable to assume that if an item was ordered, at least one unit was purchased.

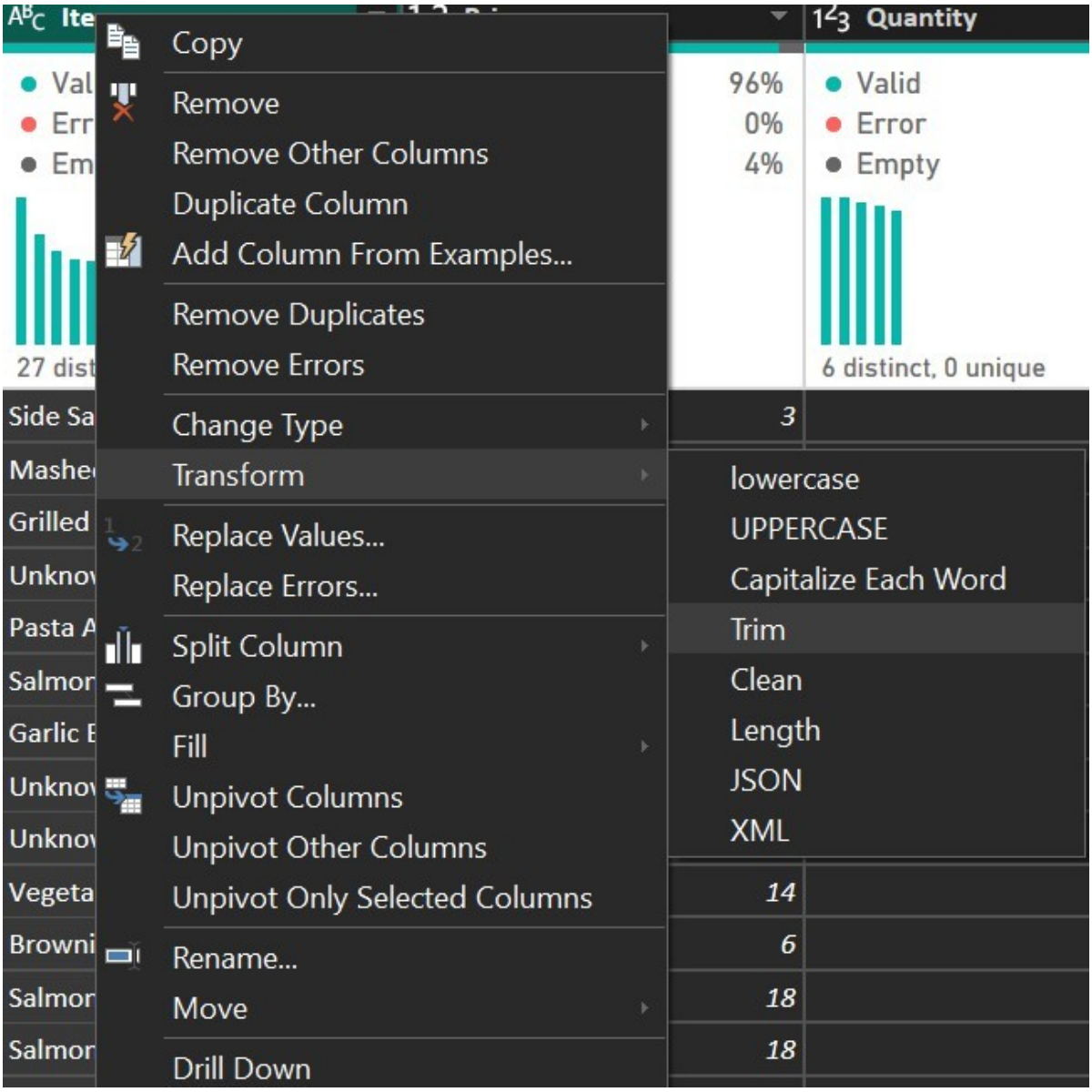


- Filling missing values

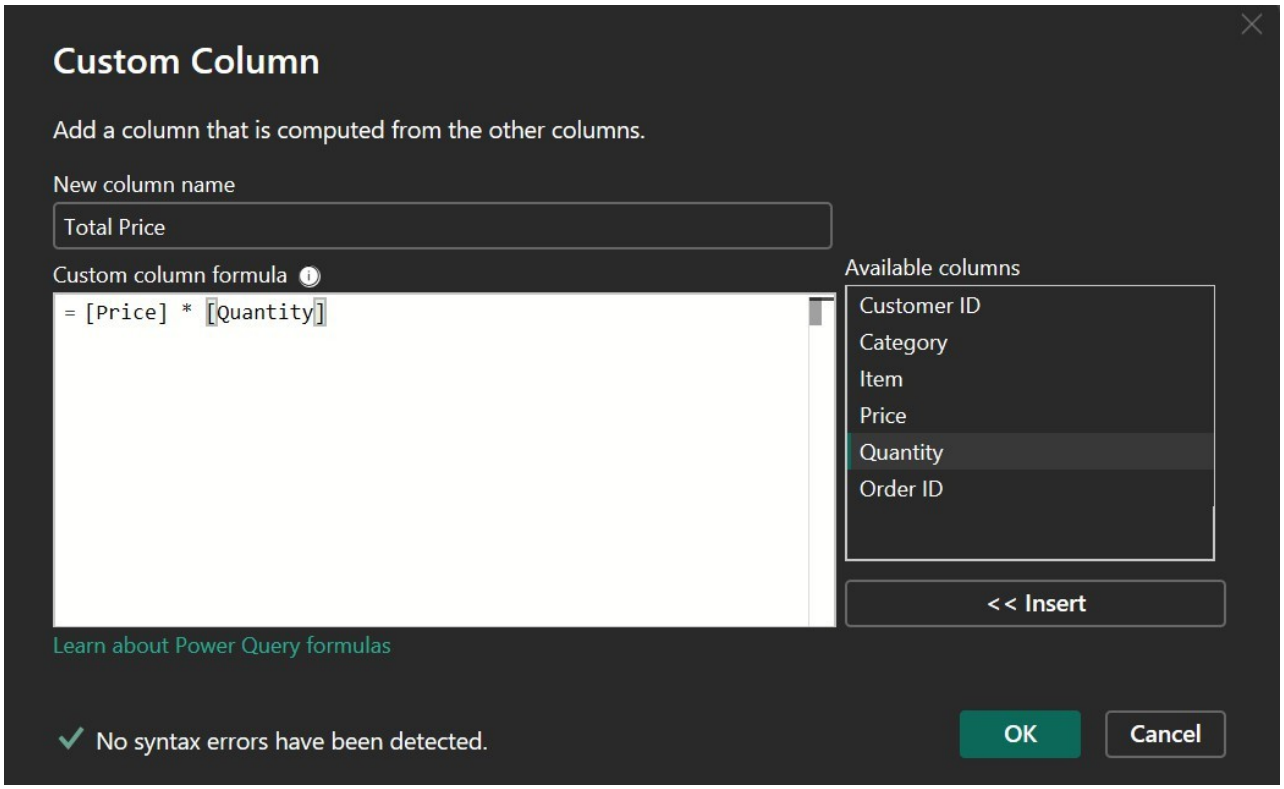


- Standardizing inconsistent data entries

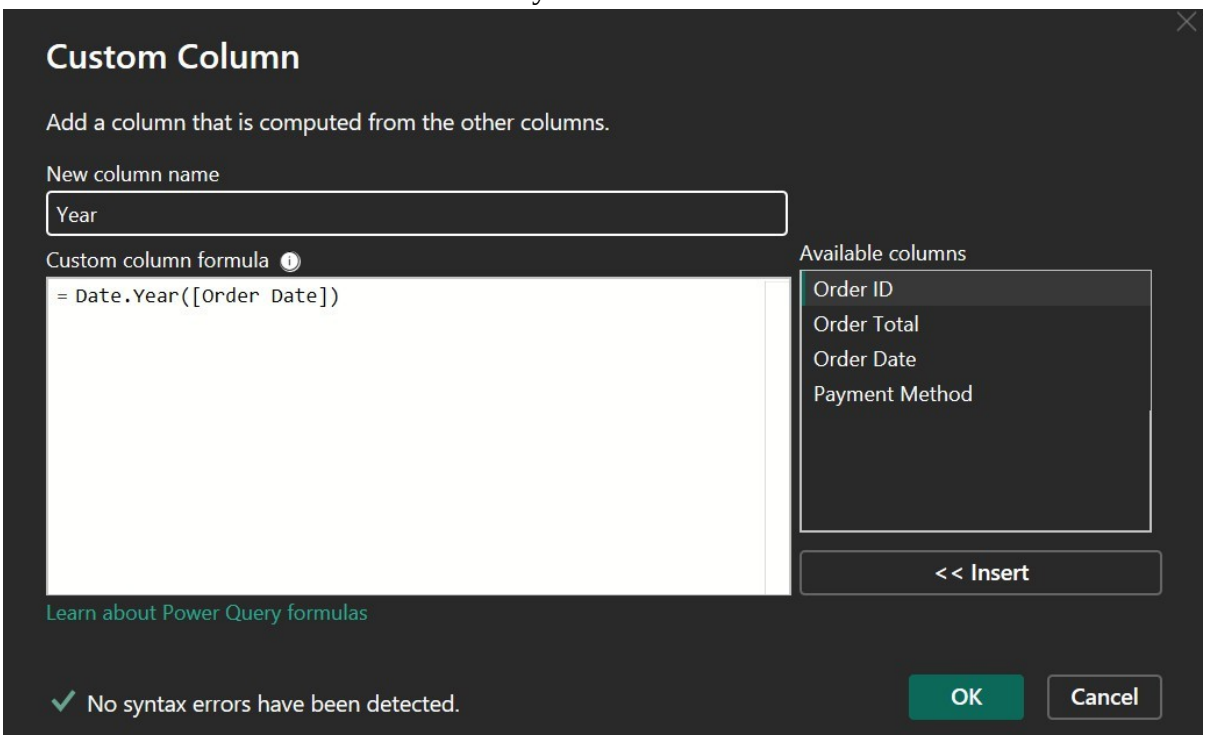
- Removed leading and trailing spaces in the Item column



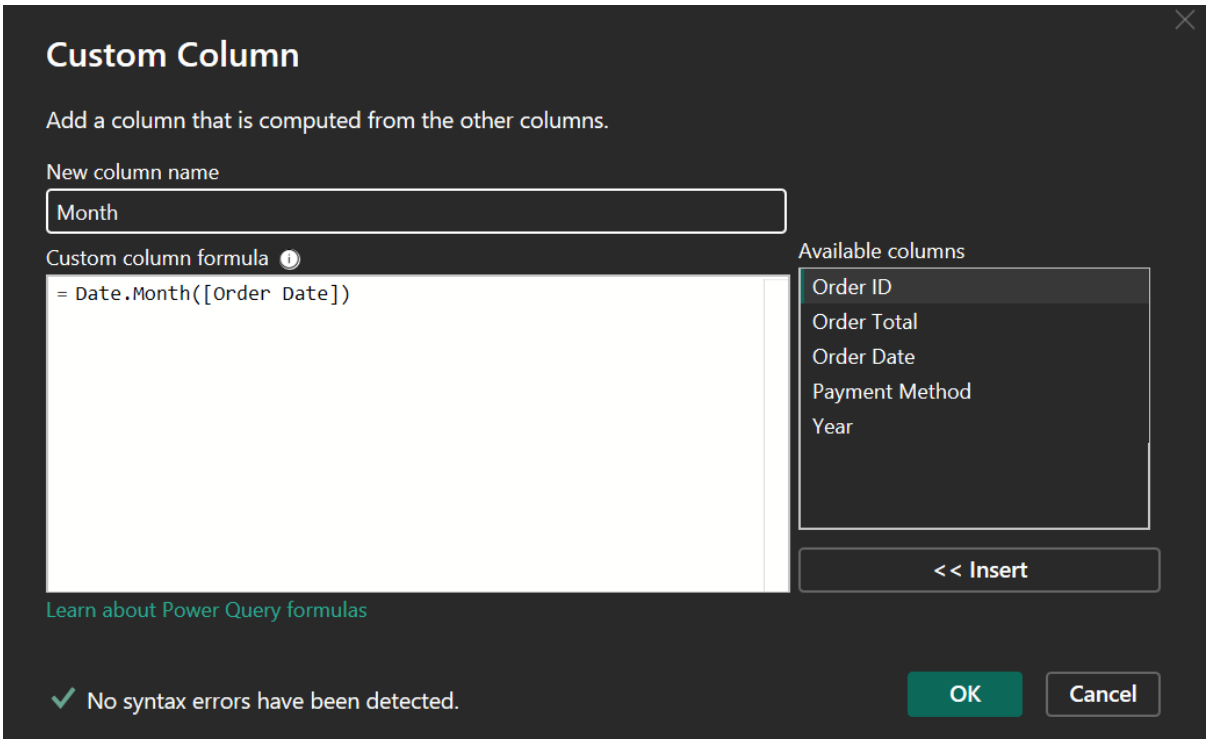
- Creating derived columns: Total Price = (Price × Quantity) in the Customer Table



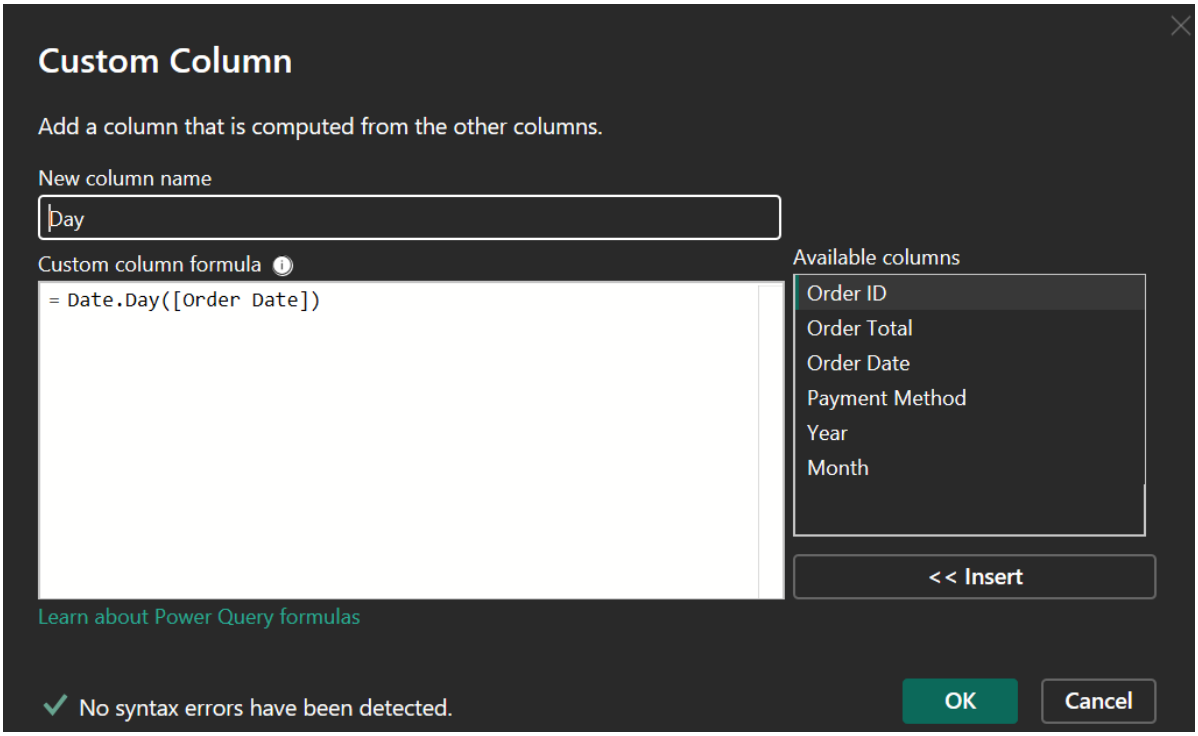
- Extracting:
  - Year from the Order Date column to analyze sales trends over time



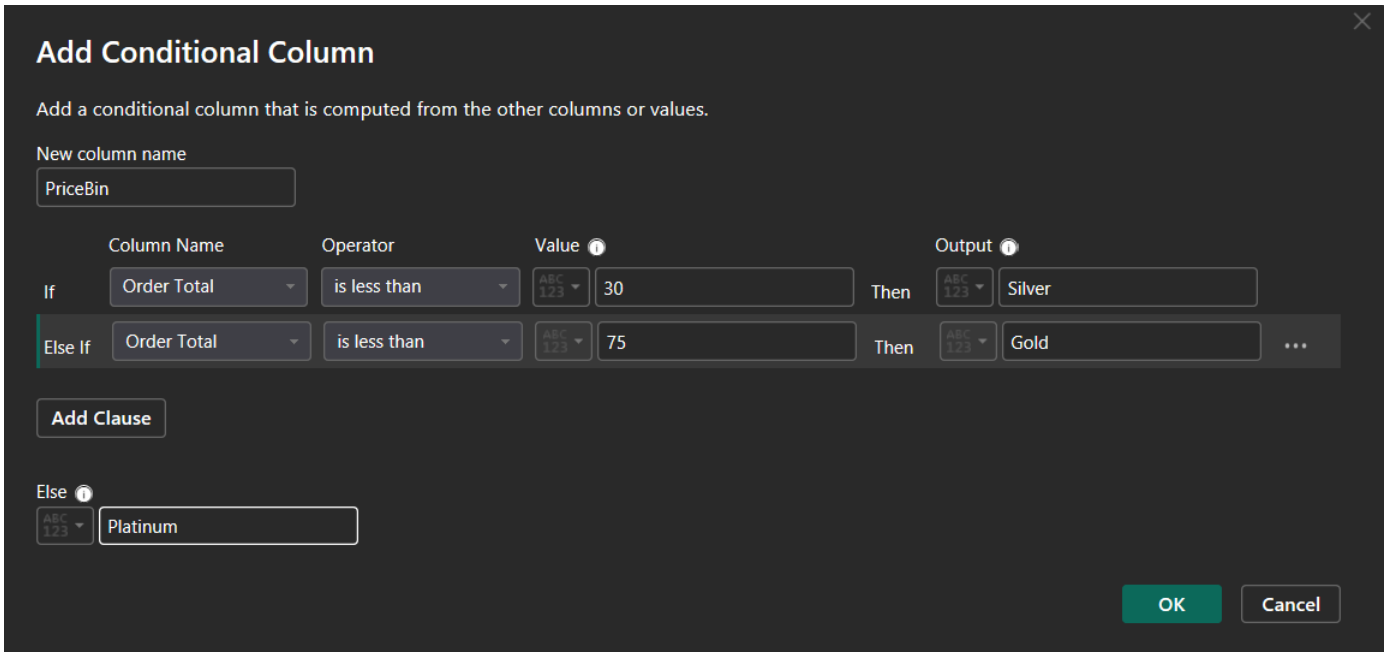
- Month from the Order Date column to identify seasonal patterns in sales



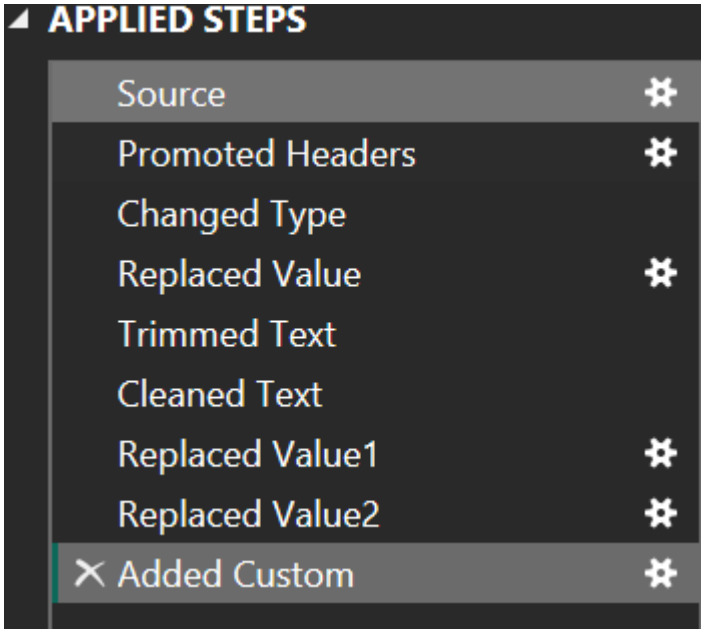
- Day of the week from the Order Date column to determine which days are busiest for the restaurant

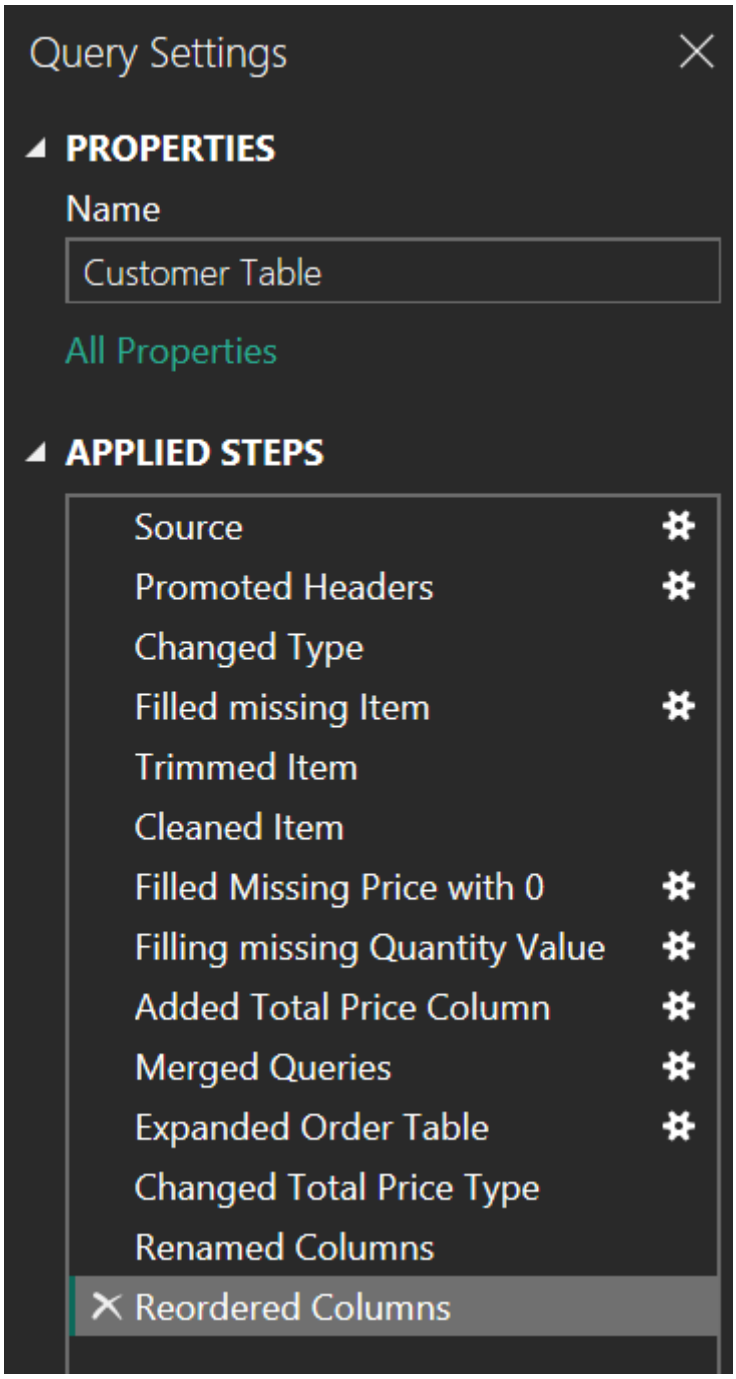


- Added price bins

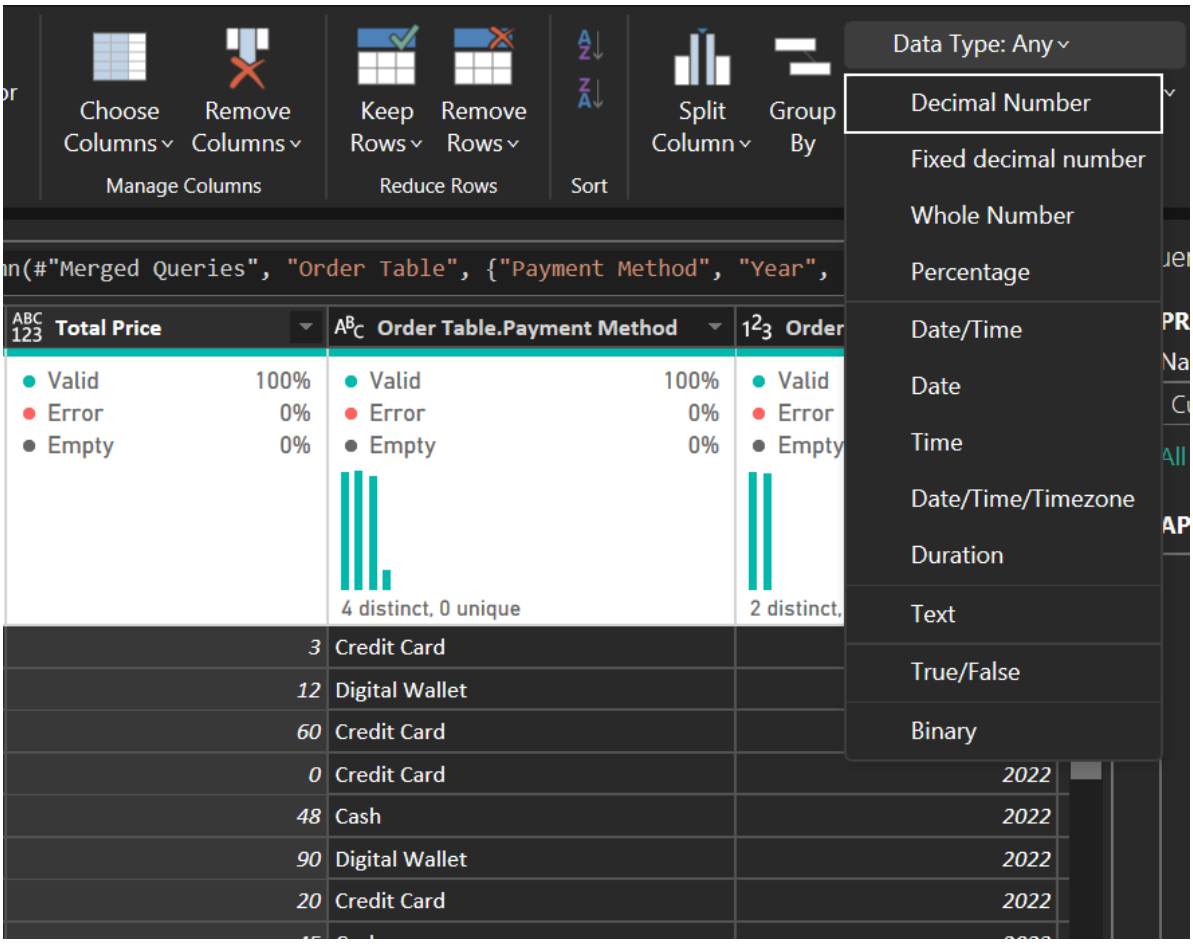


- Renaming queries professionally for better readability and organization

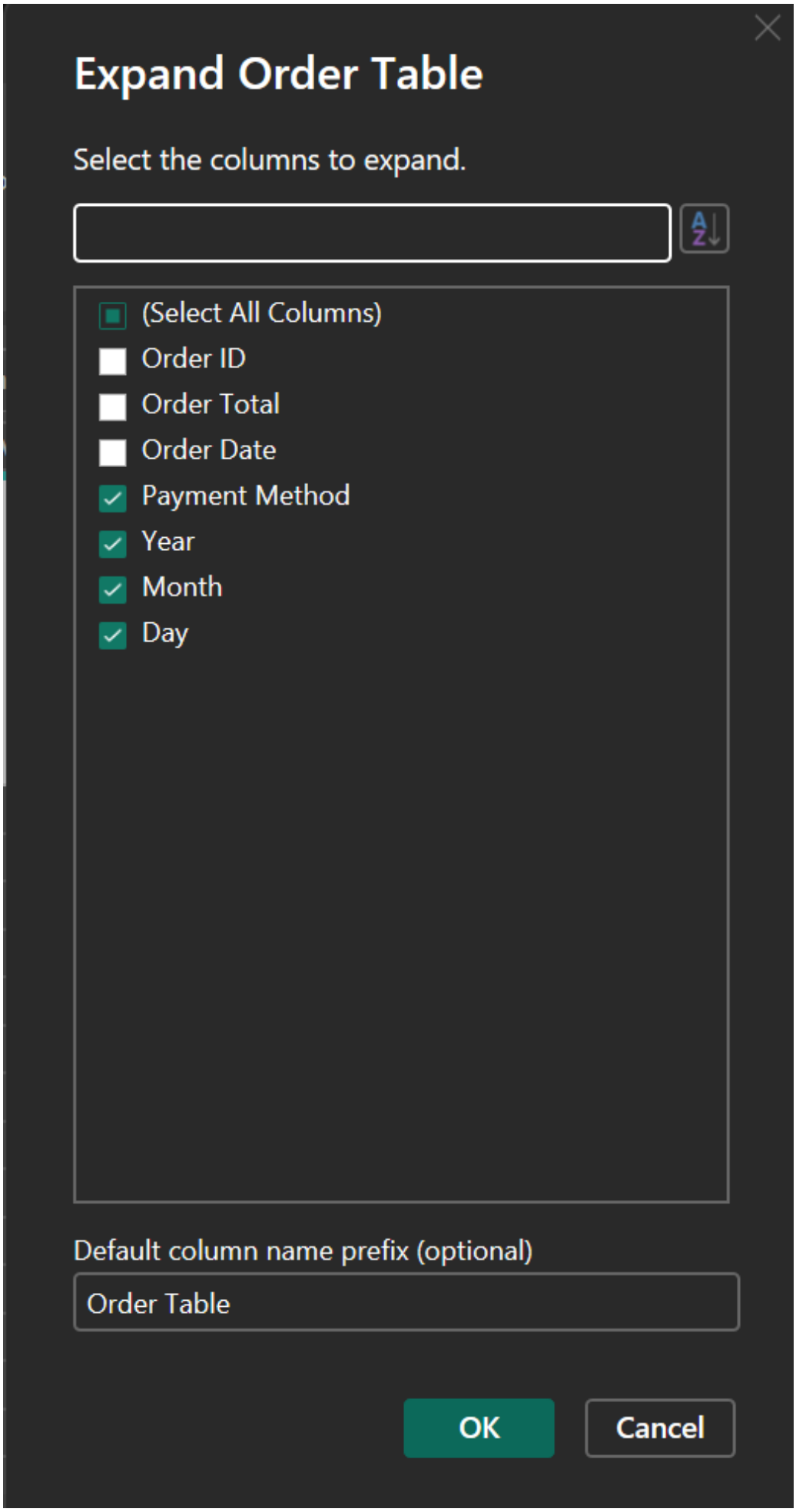
- Before Renaming 

- After Renaming 

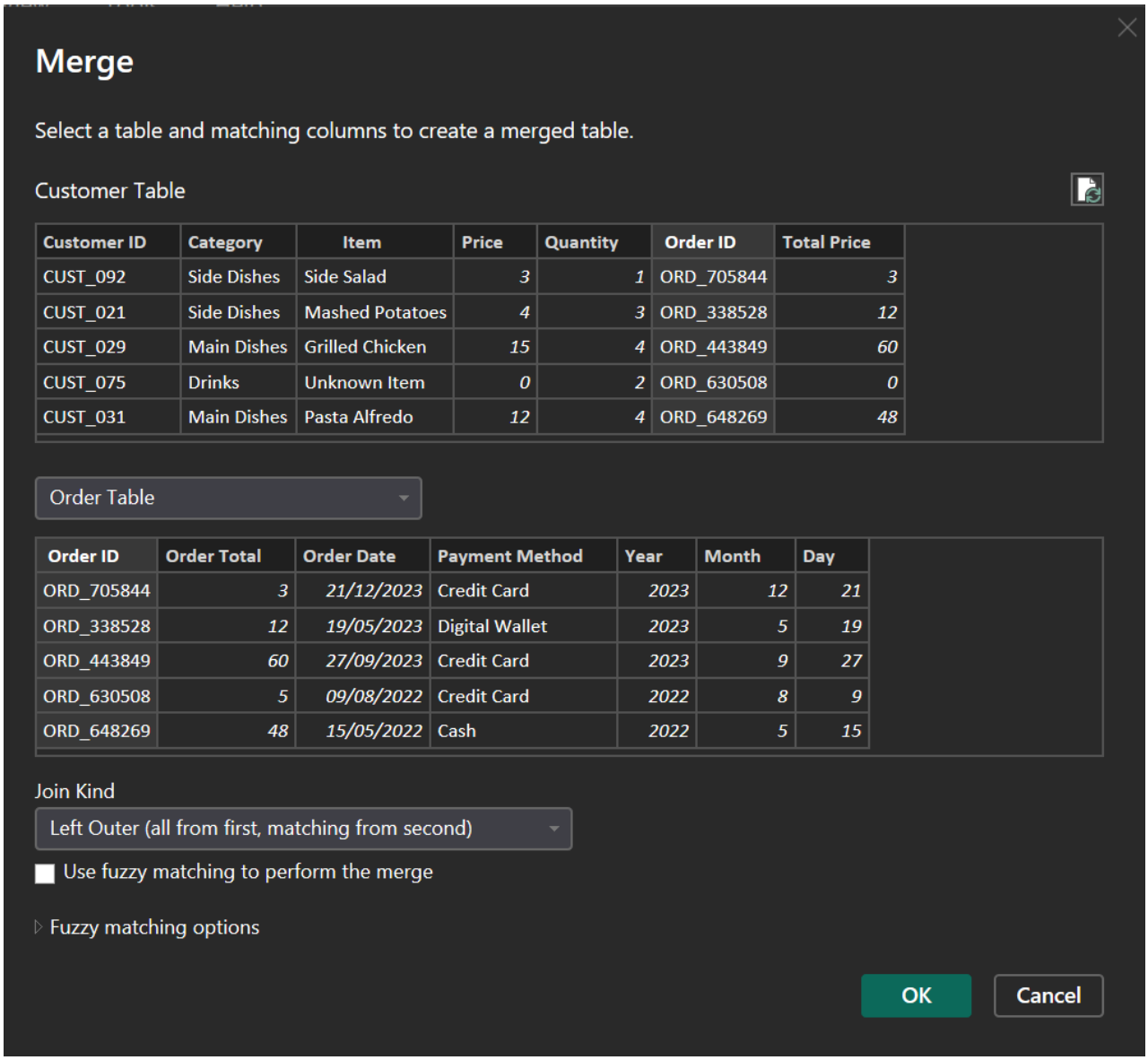
- Fixing data types:
  - Changed Total Price to Decimal Number from Any



- Removing unnecessary columns that do not contribute to the analysis



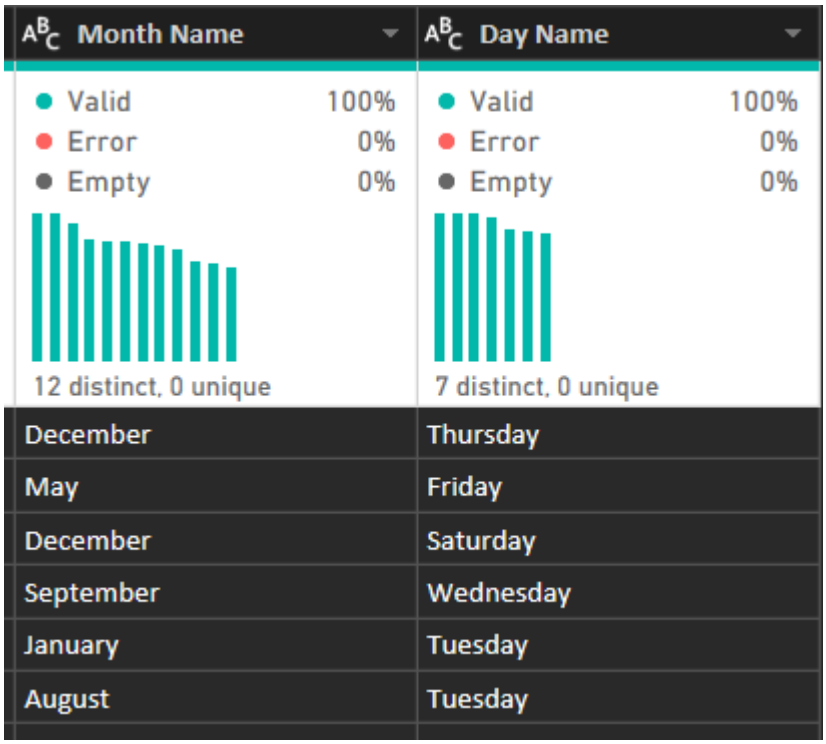
- Merging the two tables using the Order ID column to create a comprehensive dataset for analysis



Customer ID	Category	Item	Price	Quantity	Order ID	Total Price
CUST_092	Side Dishes	Side Salad	3	1	ORD_705844	3
CUST_021	Side Dishes	Mashed Potatoes	4	3	ORD_338528	12
CUST_029	Main Dishes	Grilled Chicken	15	4	ORD_443849	60
CUST_075	Drinks	Unknown Item	0	2	ORD_630508	0
CUST_031	Main Dishes	Pasta Alfredo	12	4	ORD_648269	48

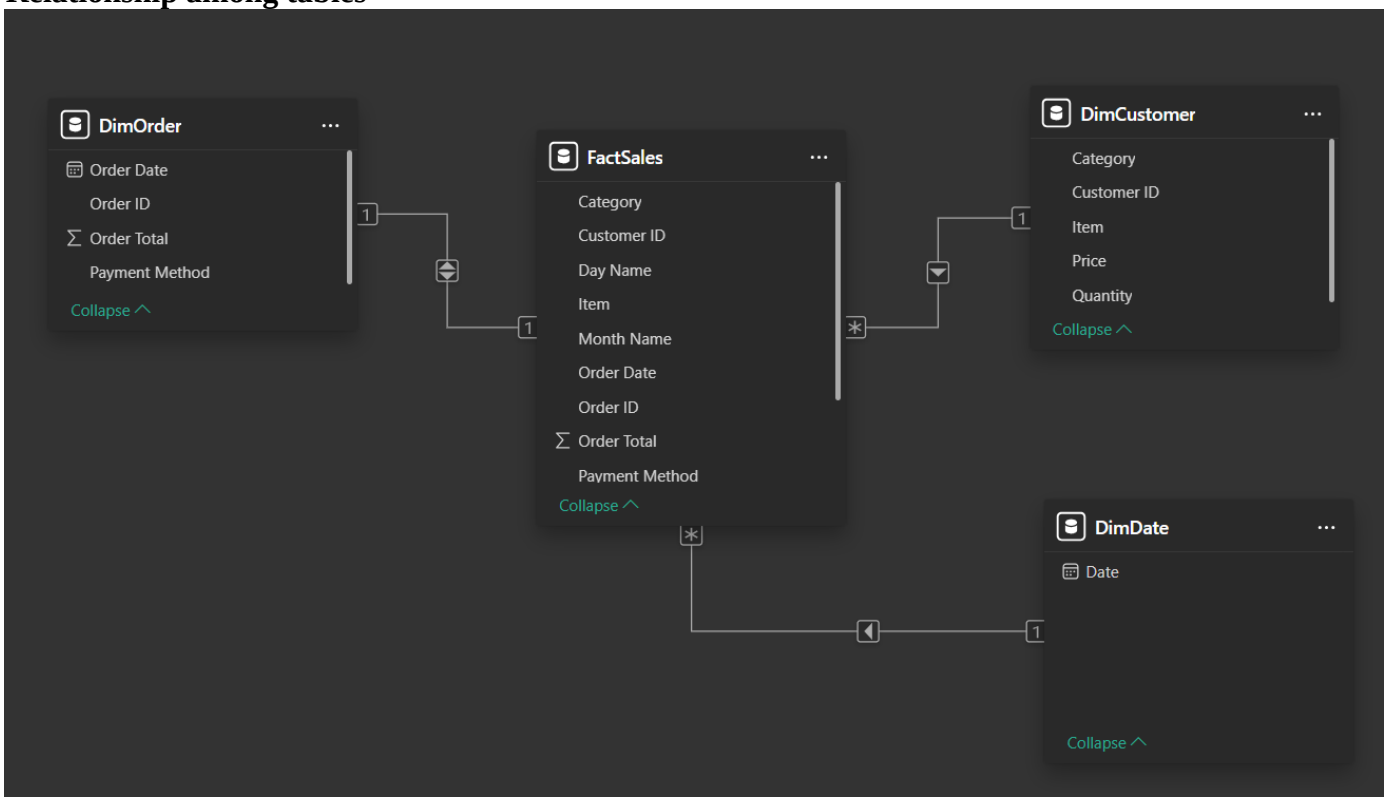
Order ID	Order Total	Order Date	Payment Method	Year	Month	Day
ORD_705844	3	21/12/2023	Credit Card	2023	12	21
ORD_338528	12	19/05/2023	Digital Wallet	2023	5	19
ORD_443849	60	27/09/2023	Credit Card	2023	9	27
ORD_630508	5	09/08/2022	Credit Card	2022	8	9
ORD_648269	48	15/05/2022	Cash	2022	5	15

- Extracted Month Name and Day Name - For time analysis.



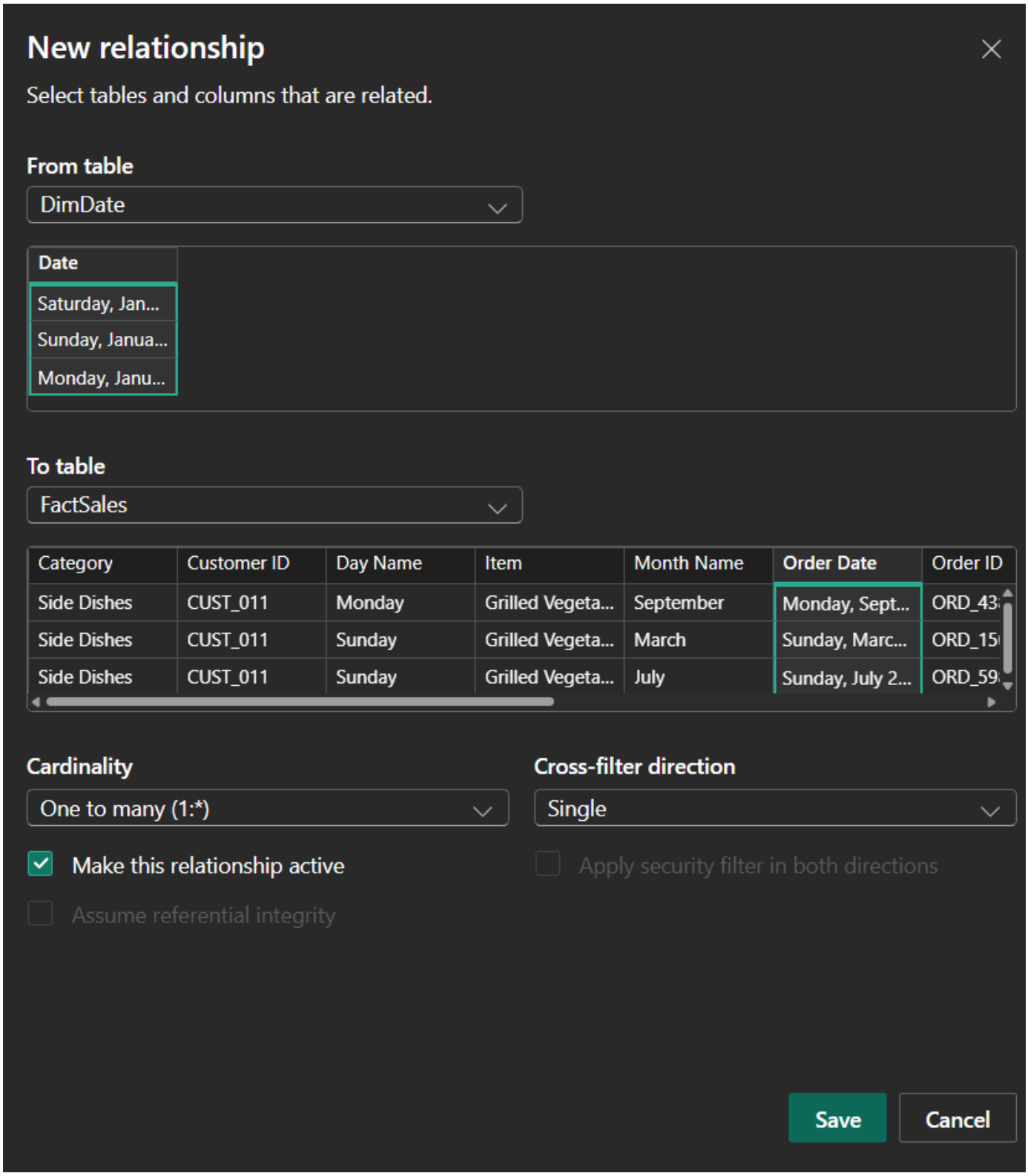
Part C: Data Modeling (15 Marks)

- **Relationship among tables**



- Creating a DateTable

```
DimDate = CALENDAR(MIN(FactSales[Order Date]), MAX(FactSales[Order Date]))
```



Identify and Justify Tables

- **Fact Table:** FactSales. Justification: It stores the measurable, quantitative transactional data (Quantity, Price, Order Total) and the foreign keys required to link to the descriptive tables.
- **Dimension Tables:** DimCustomer, DimProduct, DimDate. Justification: They contain the descriptive text attributes and time periods used to filter, slice, and group the metrics in the fact table.

Relationship Description

- DimCustomer [Customer ID] to FactSales [Customer ID]: 1-to-many (1:*), single cross-filter direction.
- DimProduct [Item] to FactSales [Item]: 1-to-many (1:*), single cross-filter direction.
- DimDate [Date] to FactSales [Order Date]: 1-to-many (1:*), single cross-filter direction.

Efficient Reporting Structure
This model utilizes a star schema optimized for the Power BI DAX engine. The single-directional 1-to-many relationships ensure filters flow downhill from dimensions to facts without creating circular dependencies. This structure minimizes data redundancy, compresses the file size, and significantly accelerates query performance for dashboard visuals.

Part D: DAX Measures and Calculated Columns (20 Marks)

1) Calculated columns

I. Transaction Size (Conditional Logic)

- **DAX Formula:** Transaction Size = SWITCH(TRUE(), FactSales[Order Total] < 20, "Small", FactSales[Order Total] <= 50, "Medium", "Large")
- **What it calculates:** Evaluates each transaction's total and categorizes it into "Small", "Medium", or "Large" buckets.
- **Decision-making value:** Allows management to analyze purchasing behaviors and see if the restaurant relies more on high-volume/low-ticket sales or low-volume/high-ticket sales, which informs promotional pricing strategies.

```
Transaction Size = SWITCH(TRUE(), FactSales[Order Total] < 20, "Small", FactSales[Order Total] <= 50, "Medium", "Large")
```

II. Is Weekend (Conditional Logic)

- **DAX Formula:** Is Weekend = IF(WEEKDAY(FactSales[Order Date], 2) >= 6, "Weekend", "Weekday")
- **What it calculates:** Looks at the order date and flags it as a "Weekend" (Saturday/Sunday) or "Weekday".
- **Decision-making value:** Helps staff scheduling and inventory planning by comparing weekend surge volume against baseline weekday traffic.

```
Is Weekend = IF(WEEKDAY(FactSales[Order Date], 2) >= 6, "Weekend", "Weekday"
```

2) DAX Measures

1. Total Revenue (Total Amount)

- **DAX Formula:** Total Revenue = SUM(FactSales[Order Total])
- **What it calculates:** Sums up the total monetary value of all transactions in the current filter context.
- **Decision-making value:** This is the primary Key Performance Indicator (KPI) for the restaurant's financial health and serves as the foundational metric for most other calculations.

```
Total Revenue = SUM(FactSales[Order Total])
```

2. Total Orders (Distinct Count)

- **DAX Formula:** Total Orders = DISTINCTCOUNT(FactSales[Order ID])
- **What it calculates:** Counts the unique number of Order IDs, ensuring that an order with multiple items is only counted as a single customer transaction.

- **Decision-making value:** Tracks overall restaurant foot traffic and transaction volume independently of how much money was spent.

```
Total Orders = DISTINCTCOUNT(FactSales[Order ID])
```

3. Average Order Value / AOV (Average Metric)

- **DAX Formula:** Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)
- **What it calculates:** Divides the total revenue by the distinct number of orders.
- **Decision-making value:** Crucial for measuring upselling success. If AOV increases, staff are successfully encouraging customers to add sides, drinks, or desserts to their main meals.

```
Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)
```

4. YTD Revenue (Time Intelligence)

- **DAX Formula:** YTD Revenue = TOTALYTD([Total Revenue], DimDate[Date])
- **What it calculates:** Calculates the cumulative total revenue from January 1st to the current date in the filter context.
- **Decision-making value:** Allows ownership to track performance against annual financial targets and compare cumulative growth against previous years.

```
YTD Revenue = TOTALYTD([Total Revenue], DimDate[Date])
```

5. Previous Month Revenue (Time Intelligence Helper)

- **DAX Formula:** Previous Month Revenue = CALCULATE([Total Revenue], DATEADD(DimDate[Date], -1, MONTH))
- **What it calculates:** Retrieves the Total Revenue for the exact same period one month prior.
- **Decision-making value:** Necessary for calculating month-over-month growth. It provides a baseline for short-term historical comparison.

```
Previous Month Revenue = CALCULATE([Total Revenue], DATEADD(DimDate[Date], -1, MONTH))
```

6. MoM Revenue Growth % (Change Over Time)

- **DAX Formula:** MoM Revenue Growth % = DIVIDE([Total Revenue] - [Previous Month Revenue], [Previous Month Revenue], 0)
- **What it calculates:** The percentage increase or decrease in revenue compared to the previous month.
- **Decision-making value:** Acts as an early warning system. Negative growth alerts management to investigate recent menu changes, seasonal dips, or operational issues.

```
MoM Revenue Growth % = DIVIDE([Total Revenue] - [Previous Month Revenue], [Previous Month Revenue], 0)
```

7. % of Total Revenue (Percentage Contribution)

- **DAX Formula:** % of Total Revenue = DIVIDE([Total Revenue], CALCULATE([Total Revenue], REMOVEFILTERS(DimProduct)))
- **What it calculates:** Calculates the revenue of a specific filter context (like a product category) and divides it by the grand total revenue across all products.

- **Decision-making value:** Highlights product dependency. It tells management exactly which menu categories (e.g., Main Dishes vs. Desserts) carry the business financially.

```
% of Total Revenue = DIVIDE([Total Revenue], CALCULATE([Total Revenue], REMOVEFILTERS(DimOrder)))
```

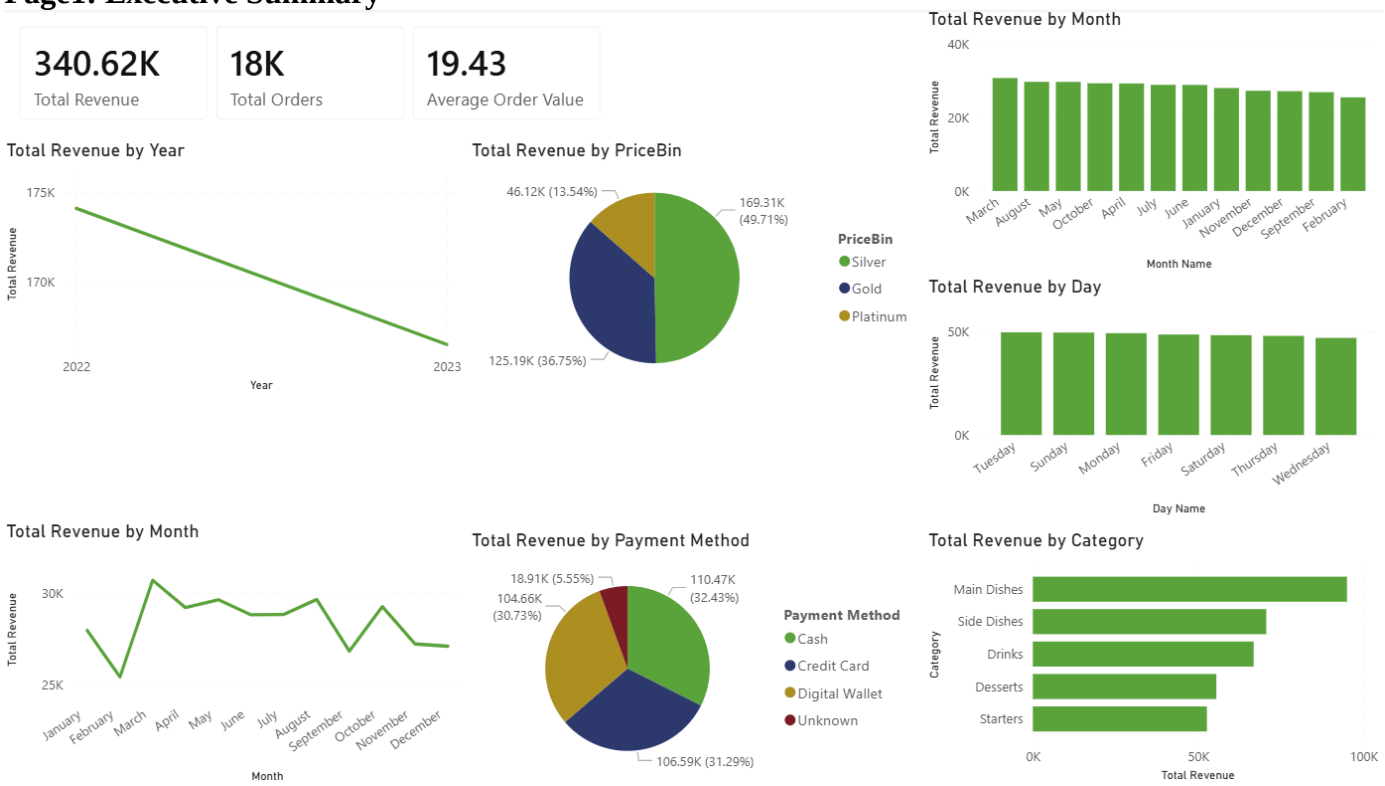
8. Product Category Rank (Ranking)

- **DAX Formula:** Category Rank = RANKX(ALL(DimProduct[Category]), [Total Revenue], , DESC, Dense)
- **What it calculates:** Assigns a numeric rank (1, 2, 3...) to product categories based on their total revenue generation.
- **Decision-making value:** Quickly identifies the "star" categories for marketing focus and the "dog" categories that might need menu redesign or removal.

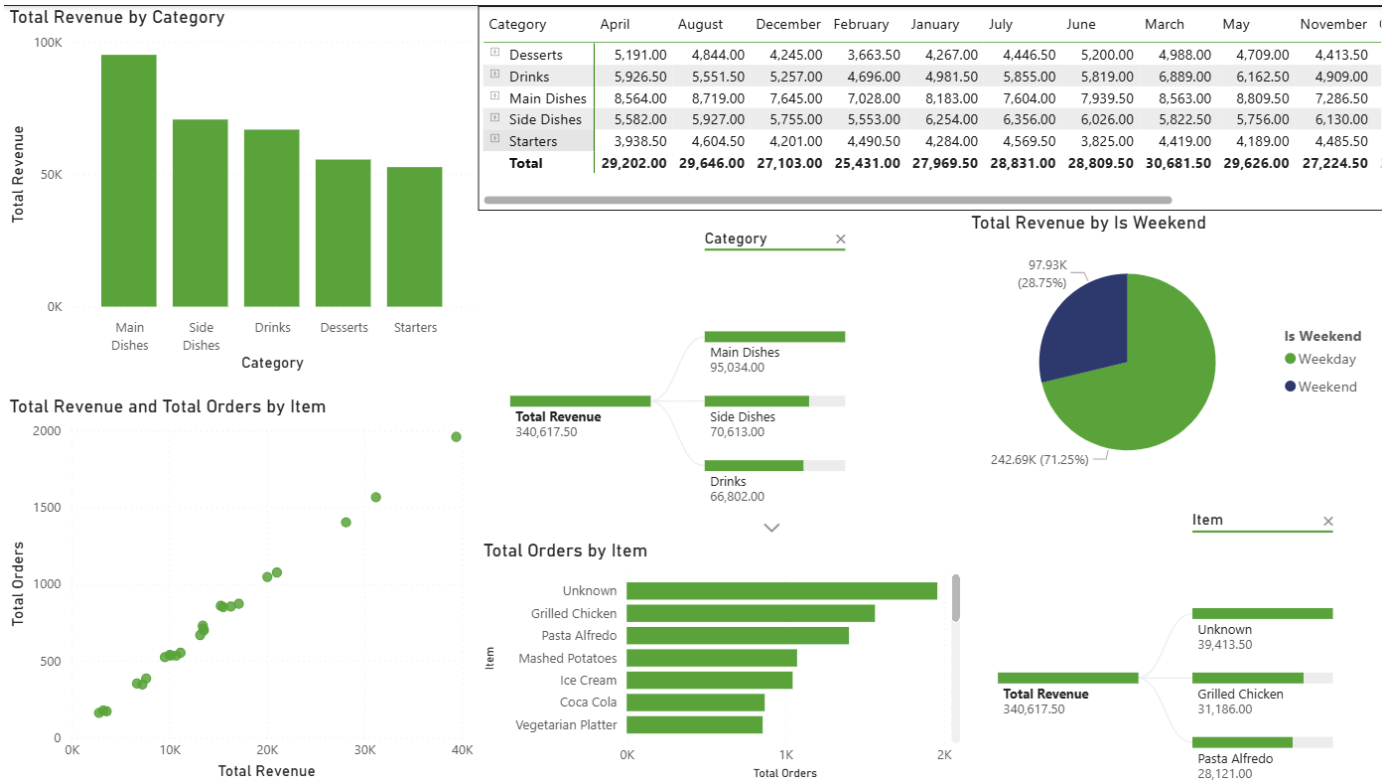
```
Category Rank = RANKX(ALL(DimCustomer[Category]), [Total Revenue], , DESC, Dense)
```

Part E: Dashboard Design and Visualization

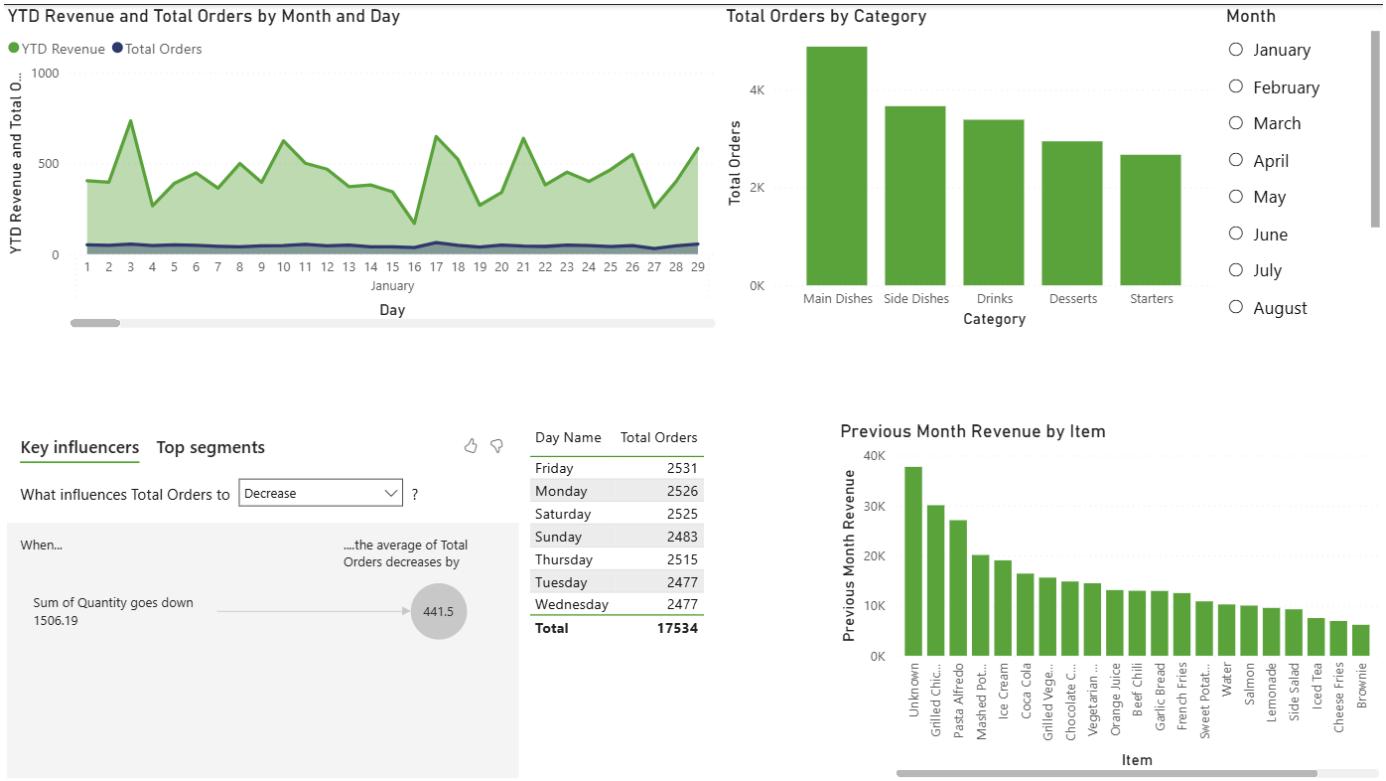
Page1: Executive Summary



Page2: Detailed Analysis



Page3: Insights and Perfomance Monitoring



Explanation of Design Choices

The dashboard employs a top-down, left-to-right information architecture aligning with natural reading patterns. High-level KPI cards are anchored at the top for immediate visibility. A minimalist, high-contrast corporate color theme is applied consistently across all three pages to reduce cognitive load. Visual clutter is minimized by removing unnecessary gridlines and data labels, relying on tooltips for exact figures. Standard business visuals (bar, line, and matrix charts) were selected over complex or 3D visuals to ensure immediate interpretability by non-technical stakeholders.

Explanation of User Interactions

The dashboard supports multi-layered data exploration through synchronized cross-filtering. Selecting a specific category, such as 'Main Dishes', in a bar chart automatically filters all other visuals on the page to reflect only that category's metrics. Global slicers for Date and Payment Method are synced across all pages to maintain context during navigation. The Detailed Analysis page features hierarchical drill-down capabilities, allowing users to expand category-level data down to individual item performance. Users can hover over any visual element to trigger native tooltips displaying exact quantitative values and context.

Part F: Analytical Insights and Business Recommendations

Analytical Insights

Insight 1: Revenue Decline. Total Revenue dropped from approximately 175K in 2022 to roughly 166K in 2023.

Insight 2: Flat Daily Demand. Revenue is evenly distributed across all days of the week. Weekdays account for 71.25% of total revenue, and weekends account for exactly 28.75%.

Insight 3: Category Dominance. "Main Dishes" is the primary revenue driver, generating over 95K. This significantly outperforms "Starters", which generated approximately 50K.

Insight 4: Item-Level Concentration and Data Quality. Within Main Dishes, revenue is heavily concentrated in just two items: Grilled Chicken (31,186) and Pasta Alfredo (28,121). "Unknown" items account for an even larger share of overall revenue (39,413.50).

Insight 5: Payment Method Parity. Customers exhibit no strong preference for payment methods. Cash (32.43%), Credit Card (31.29%), and Digital Wallet (30.73%) are used almost equally.

Actionable Business Recommendations

Recommendation 1: Investigate Data Capture Issues. The high revenue attributed to "Unknown" items (39,413.50) indicates a critical failure in the data. Management must immediately audit the Point of Sale (POS) system and staff data entry protocols to accurately classify these sales and understand true product performance.

Recommendation 2: Implement Weekend Promotions. Daily revenue is entirely flat across the week. Weekends show no natural surge in customer spending compared to weekdays. The restaurant should introduce weekend-exclusive bundles, live entertainment, or events to drive higher foot traffic on Saturdays and Sundays.

Recommendation 3: Menu Diversification. "Main Dishes" relies heavily on just Grilled Chicken and Pasta Alfredo. This exposes the business to supply chain disruptions for these specific ingredients. Introduce new premium main dishes to spread revenue generation across a wider variety of menu items.

Conclusion

This project successfully delivered a comprehensive Business Intelligence solution to analyze and optimize restaurant sales performance. By ingesting and transforming over 17,000 raw transactional records into a structured star schema, we established a robust foundation for advanced analytics. The deployment of DAX-driven metrics and an interactive, three-page dashboard provided immediate visibility into critical operational areas. The analysis revealed significant revenue concentration in specific main dishes, flat daily demand, and critical data capture issues at the point of sale. By acting on the data-driven recommendations—specifically implementing strategic menu bundling, launching weekend promotions, and auditing POS data entry—the restaurant management can mitigate supply chain risks, increase average order value, and drive sustainable revenue growth.

Part G: GitHub Repository and Documentation (5 Marks)

1) GitHub Link: https://github.com/SamanthaMasaki/PowerBI_EndSemester_Project
2) Kaggle Link: https://www.kaggle.com/datasets/ahmedmohamed2003/restaurant-sales-dirty-data-for-cleaning-training